

WELL COMPLETION REPORT (SYNTHETIC)

OIL-LKW-002 - Lakwa-002

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000016
Field / District	Lakwa / Charaideo
Block	PEL-ASSAM-04
Basin	Assam-Arakan
Formation	Kopili (KOP)
Spud / Completion	2011-06-18 / 2011-11-16
Measured / TVD	3076 m / 3047 m
Trajectory	Deviated; inc 14.8 deg; az 64.6 deg
Formation Interval	1374-2148 m
Lithology	Compact shale with sandstone streaks
Pore Pressure Reference	2746 psi
Fracture Gradient	15.33 ppg
Mud System	KCl Polymer WBM
NWIS Well Risk	Moderate (55.4/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	769	Surface
9 5/8 in casing	1784	Intermediate
7 in liner/casing	2706	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2011-07-09 566 m	Stuck Pipe	Critical	Rotary movement deteriorated progressively. Mitigation: Apply approved freeing procedure and avoid excessive overpull. Lesson: Avoid prolonged static periods in high-drag interval.
2011-09-13 1889 m	Torque Spike	High	Torque rose without corresponding pressure anomaly. Mitigation: Monitor torque derivative rather than absolute torque alone. Lesson: Use offset torque trend to set adaptive operating envelope.
2011-09-16 1949 m	Fishing Operation	High	Mechanical obstruction prevented normal drilling progression. Mitigation: Review BHA failure mode before resuming drilling. Lesson: Maintain detailed BHA inventory for rapid recovery planning.
2011-11-05 2866 m	Pressure Spike	High	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.
2011-11-06 2894 m	Pressure Spike	High	Pressure trend exceeded recent moving baseline. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.
2011-11-08 2923 m	Stuck Pipe	High	High drag was recorded while circulating remained possible. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Avoid prolonged static periods in high-drag interval.
2011-07-12 632 m	Kick	Medium	Drilling break followed by positive flow check. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Treat pit gain plus gas increase as an early influx indicator.

Completion Summary

The synthetic well reached 3076 m MD in the Kopili interval. The NWIS knowledge record links 21 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.